

2. TEST CASE ANALYSIS REPORT

Wipro Training Capstone Automation Project

This document contains a comprehensive set of test cases covering all functional modules of the banking application under test. Each test case specifies the module, test scenario, preconditions, test data, steps to execute, expected result, and priority level. These test cases are designed to validate both positive and negative flows of the application.

2.1 Module Overview

Module	Test Cases	Scenarios Covered
Login	4	Valid login, Invalid login, Empty username, Empty password
Customer	3	Add customer
Account	3	Create account
Transaction	5	Deposit, Withdraw, Fund transfer, Invalid transfer
Security	2	Logout
TOTAL	20	Comprehensive coverage across 6 modules

2.2 Login Module

The Login Module validates user authentication. It covers valid credentials, invalid credentials, and boundary conditions such as empty username or password fields.

TC_LOGIN_001 - Valid Login

Field	Value
Test Case ID	TC_LOGIN_001
Module	Login
Scenario	Valid Login
Preconditions	User is on the Login page; application is running
Test Data	Username: mng662888 Password: nezyUm
Steps to Execute	1. Navigate to Login page 2. Enter valid username 3. Enter valid password 4. Click Login button
Expected Result	User is redirected to the Home page successfully
Priority	High
Status	To Be Executed

TC_LOGIN_002 - Invalid Login

Field	Value
Test Case ID	TC_LOGIN_002
Module	Login

Field	Value
Scenario	Invalid Login with incorrect credentials
Preconditions	User is on the Login page
Test Data	Username: invalidUser Password: wrongPass
Steps to Execute	1. Navigate to Login page 2. Enter invalid username 3. Enter invalid password 4. Click Login button
Expected Result	An error alert/message is displayed: 'Invalid username or password'. User remains on the Login page.
Priority	High
Status	To Be Executed

TC_LOGIN_003 - Empty Username

Field	Value
Test Case ID	TC_LOGIN_003
Module	Login
Scenario	Login attempt with empty username field
Preconditions	User is on the Login page
Test Data	Username: (empty) Password: eguvenA
Steps to Execute	1. Navigate to Login page 2. Leave username field blank 3. Enter a valid password 4. Click Login button
Expected Result	Validation error message is displayed: 'User or Password is not valid '. Login is not attempted.
Priority	Medium
Status	To Be Executed

TC_LOGIN_004 - Empty Password

Field	Value
Test Case ID	TC_LOGIN_004
Module	Login
Scenario	Login attempt with empty password field
Preconditions	User is on the Login page
Test Data	Username: manager Password: (empty)
Steps to Execute	1. Navigate to Login page 2. Enter a valid username 3. Leave password field blank 4. Click Login button
Expected Result	Validation error message is displayed: ' User or Password is not valid '. Login is not attempted.
Priority	Medium
Status	To Be Executed

2.3 Customer Module

The Customer Module covers CRUD operations on customer records. Test cases verify successful creation with auto-generated Customer IDs, editing of existing records, and safe deletion.

TC_CUST_001 - Add New Customer

Field	Value
Test Case ID	TC_CUST_001
Module	Customer
Scenario	Add a new customer to the system
Preconditions	User is logged in with appropriate role/permissions
Test Data	Name: Shahid DoB: 06/11/2001 Address: Jamshedpur Email: shahid@gmail.com Mobile: 9876543210
Steps to Execute	1. Navigate to New Customer page 2. Fill in all mandatory fields 3. Click Submit
Expected Result	Customer is created successfully. A unique Customer ID is auto-generated and displayed. Success confirmation message shown.
Priority	High
Status	To Be Executed

2.4 Account Module

The Account Module handles bank account lifecycle management including creation, modification, and deletion. An Account must be linked to an existing Customer.

TC_ACC_001 - Create Account

Field	Value
Test Case ID	TC_ACC_001
Module	Account
Scenario	Create a new bank account for an existing customer
Preconditions	User is logged in; valid Customer ID exists
Test Data	Customer ID: (existing) Account Type: Savings Initial Deposit: 5000
Steps to Execute	1. Navigate to New Account page 2. Enter Customer ID 3. Select Account Type 4. Enter initial deposit 5. Click Submit
Expected Result	Account is created successfully. A unique Account ID is auto-generated and displayed. Confirmation message shown.
Priority	High
Status	To Be Executed

2.5 Transaction Module

The Transaction Module covers all financial operations including deposits, withdrawals, and fund transfers. Negative scenarios such as insufficient balance and invalid account transfers are also validated.

TC_TRANS_001 - Deposit Money

Field	Value
Test Case ID	TC_TRANS_001
Module	Transaction
Scenario	Deposit money into a valid account
Preconditions	User is logged in; valid Account ID exists
Test Data	Account ID: (existing) Deposit Amount: 1000
Steps to Execute	1. Navigate to Deposit page 2. Enter Account ID 3. Enter deposit amount 4. Click Submit
Expected Result	Transaction is successful. Account balance is updated and reflects the deposited amount. Transaction confirmation/receipt is displayed.
Priority	High
Status	To Be Executed

TC_TRANS_002 - Withdraw Money

Field	Value
Test Case ID	TC_TRANS_002
Module	Transaction
Scenario	Withdraw money from an account with sufficient balance
Preconditions	User is logged in; Account exists; Balance $\geq$ withdrawal amount
Test Data	Account ID: (existing) Withdrawal Amount: 500
Steps to Execute	1. Navigate to Withdrawal page 2. Enter Account ID 3. Enter withdrawal amount 4. Click Submit
Expected Result	Transaction is successful. Account balance is reduced by the withdrawn amount. Transaction confirmation is displayed.
Priority	High
Status	To Be Executed

TC_TRANS_003 - Insufficient Balance Withdrawal

Field	Value
Test Case ID	TC_TRANS_003
Module	Transaction
Scenario	Withdraw amount exceeding available balance
Preconditions	User is logged in; Account balance is less than withdrawal amount
Test Data	Account ID: (existing) Withdrawal Amount: greater than current balance

Field	Value
Steps to Execute	1. Navigate to Withdrawal page 2. Enter Account ID 3. Enter amount exceeding balance 4. Click Submit
Expected Result	Error message is displayed: 'Insufficient balance'. Transaction is not processed. Account balance remains unchanged.
Priority	High
Status	To Be Executed

TC_TRANS_004 - Fund Transfer

Field	Value
Test Case ID	TC_TRANS_004
Module	Transaction
Scenario	Transfer funds between two valid accounts
Preconditions	User is logged in; both source and destination accounts exist; source has sufficient balance
Test Data	Source Account ID: (existing) Destination Account ID: (existing) Transfer Amount: 200
Steps to Execute	1. Navigate to Fund Transfer page 2. Enter source Account ID 3. Enter destination Account ID 4. Enter transfer amount 5. Click Submit
Expected Result	Transfer is completed successfully. Source account balance is debited. Destination account balance is credited. Confirmation message is shown.
Priority	High
Status	To Be Executed

TC_TRANS_005 - Invalid Account Transfer

Field	Value
Test Case ID	TC_TRANS_005
Module	Transaction
Scenario	Attempt fund transfer to a non-existent/invalid account
Preconditions	User is logged in; source account exists with sufficient balance
Test Data	Source Account ID: (existing) Destination Account ID: 99999999 (invalid)
Steps to Execute	1. Navigate to Fund Transfer page 2. Enter source Account ID 3. Enter invalid destination Account ID 4. Enter amount 5. Click Submit
Expected Result	Transfer fails. Error message is displayed: 'Invalid payee account number'. No debit occurs on the source account.
Priority	High
Status	To Be Executed

2.7 Security Module

The Security Module validates critical access control features including password management and session termination. These cases ensure application security and proper session handling.

TC_SEC_001 - Change Password

Field	Value
Test Case ID	TC_SEC_001
Module	Security
Scenario	Change the current user's login password
Preconditions	User is logged in
Test Data	Old Password: Test@1234 New Password: NewPass@5678 Confirm Password: NewPass@5678
Steps to Execute	1. Navigate to Change Password page 2. Enter current (old) password 3. Enter new password 4. Confirm new password 5. Click Submit
Expected Result	Password is changed successfully. Confirmation message is displayed. User can log in with the new password. Old password is no longer valid.
Priority	High
Status	To Be Executed

TC_SEC_002 - Logout

Field	Value
Test Case ID	TC_SEC_002
Module	Security
Scenario	User logs out of the application
Preconditions	User is currently logged in and on any page
Test Data	N/A
Steps to Execute	1. Click the Logout button/link 2. Confirm logout if prompted
Expected Result	User session is terminated. User is redirected to the Login page. Accessing any protected URL after logout redirects back to Login page (session invalidation).
Priority	High
Status	To Be Executed

2.8 Test Case Summary

Metric	Count / Detail
Total Test Cases	20
Positive Test Cases	14

Negative Test Cases	6 (TC_LOGIN_002, TC_LOGIN_003, TC_LOGIN_004, TC_TRANS_003, TC_TRANS_005, TC_SEC_001 old-pass check)
Total Modules	6 (Login, Customer, Account, Transaction, Statement, Security)
Priority - High	12
Priority - Medium	8
Automation Tool	Selenium WebDriver with Java / TestNG
Execution Status	All cases: To Be Executed

Note: All test cases are part of the Wipro Training Capstone Automation Project. Test execution will be performed using Selenium WebDriver. Results will be captured in a separate Test Execution Report.